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Roll No

CE-6005(2)-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Computational Methods in Structural Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Write the matrix formulation for the principle of virtual work and energy principles. 5
- b) Compare stiffness and flexibility method. 2
- c) Derive the structure stiffness matrix for the system shown in fig 1. 7

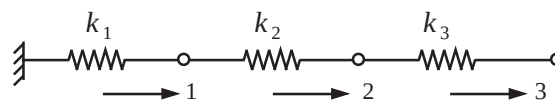


Figure 1

2. a) How the advantage of symmetry of structure and loading is taken, in matrix structural analysis? Explain with help of examples. 7
- b) Analyse the beam as shown in fig.2 by direct stiffness method (EI is constant) 20 kN/m. 7

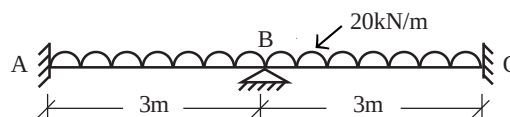


Figure 2

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[2]

3. a) Explain various storage schemes for handling the structure stiffness matrix and explain the one which you think the most suitable. 7
b) Explain clearly how will you exploit symmetry and antisymmetry in the formulation of the stiffness matrix and load vector of a structure for a given loading condition. 7
4. a) Derive the elasticity matrix for plane stress and plain strain 2D-problem. 10
b) Derive the equation for generalized element stiffness matrix. 4
5. a) Discuss about Lagrangian and Serendipity family elements in natural coordinates. 6
b) Write and draw the shape functions for first order. 4
c) Discuss about the plate bending elements. 4
6. a) Write a short note on Triangular and Rectangular elements. 5
b) Discuss about convergence requirements. 5
c) Explain about basic concepts of finite element analysis. 4
7. a) Prove that the element stiffness for a finite element is given by 10
$$[K_e] = \int_V [B]^T [D] [B] dV$$

b) Write a short note on Jacobian matrix. 4
8. Write a short note of the following: (any two) 14
 - i) State various penalty methods.
 - ii) What is direct stiffness method?
 - iii) Explain degree of freedom.

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